

Unit 1.3 Practice Packet

Program Design & Development



Practice Makes Prepared

Parts 1–2 are vocabulary that shows up directly on the AP exam — know these words cold. Parts 3–8 apply them the way the exam does. Use your guided notes and vocab sheet if you get stuck.

Part 1 • Vocab Match

Write the letter of the matching definition on each line.

- _____ 1. Iterative development
- _____ 2. Incremental development
- _____ 3. Program requirements
- _____ 4. Program specification
- _____ 5. Investigation
- _____ 6. Prototype
- _____ 7. Storyboard
- _____ 8. Module
- _____ 9. Program documentation
- _____ 10. Comments
- _____ 11. User testing
- _____ 12. Acknowledgment

- A. A sketch of each screen in order, showing what the user sees and does
- B. Documentation written in the program that is not executed
- C. Breaking a problem into smaller pieces and making sure each piece works before adding it to the whole
- D. Describe how a program functions; may include descriptions of user interactions
- E. An early, working model of a program used to test ideas
- F. A written description of the function of a code segment, event, procedure, or program and how it was developed
- G. Watching real users try a program to find problems
- H. Development that requires refinement and revision based on feedback, testing, or reflection
- I. Naming the source or original author of code you did not write
- J. Defines the requirements for a program
- K. Gathering information (surveys, interviews, observation) to find out what users need
- L. A separate part of a program that handles one job

Part 2 • Fill It In

Use each word from the word bank once.

iterative

incremental

requirements

prototype

comments

documentation

survey

acknowledge

1. Before coding, a team sends a _____ to 500 students asking which features they want.
2. A rough version of an app built to try out ideas with users is a _____.
3. "The app must let users log in with their school ID" is one of the program's _____.
4. Lines starting with # that explain the code but are not run are _____.
5. Changing a design after testers say it is confusing is _____ development.
6. Building the login screen, testing it, then adding the next screen is _____ development.
7. When you use code a classmate wrote, you must _____ them in your program.
8. A written guide explaining what each procedure does and how the program was built is program _____.

Name: _____

Class: _____

Date: _____

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Iterative, Incremental & the Process

Part 3 · Iterative, Incremental, or Neither?

Check the best label for each. **Iterative** = revise based on feedback or testing. **Incremental** = add one working piece at a time.

Scenario	Iterative	Incremental	Neither
1. A team finishes the scoring system and tests it before adding the leaderboard.			
2. After beta testers say the text is too small, the team makes it bigger.			
3. A developer adds a comment explaining a tricky loop.			
4. A chat app adds emoji reactions only after basic messaging works and is tested.			
5. Survey results show users want dark mode, so the team revises its color design.			
6. A programmer rewrites a procedure after a test case fails.			
7. A student copies an entire project from the Internet.			
8. An online store builds and tests the product page, then the cart, then checkout.			

Part 4 · Which Phase?

Write **Investigate**, **Design**, **Prototype**, or **Test**.

Activity	Phase
1. Interviewing teachers about how they take attendance	
2. Sketching each screen of the app on paper	
3. Building a basic clickable version where only the main button works	
4. Entering boundary values like 0 and 100 to check the grading code	
5. Watching students use the library checkout system	
6. Splitting the program into login, menu, and payment modules	

Part 5 · Fix It!

```
# stuff
total ← 0
FOR EACH p IN prices
{ total ← total + p }
# copied from internet
tax ← total * 0.08
```

(a) Why are these comments poor documentation?

(b) Rewrite both comments so they are useful.

Name: _____

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Analyze a Development Story

Part 6 • Case Study

Ms. Rivera's class is building a **Lost & Found app** for their school.

1. They interview the front-office staff and send a survey to 150 students.
2. They decide the app *must* let students post a photo of a found item and search by category.
3. They sketch every screen on paper.
4. They build the posting feature and test it before they start on search.
5. In a test with 10 students, users can't find the search button, so the team moves it to the top of the screen.
6. One teammate uses a photo-upload procedure from an online tutorial.

Part 6 • Questions

a) Which investigation methods did they use?

b) Identify one program requirement.

c) Which step shows **incremental** development? Explain.

d) Which step shows **iterative** development? Explain.

e) Step 3 is part of which phase? What is this sketch called?

f) What must the teammate in step 6 do?

Part 7 • Choose the Method

Pick the best investigation method: **survey**, **interview**, **user testing**, or **direct observation**. Explain why.

Goal	Method	Why?
1. Learn how 400 students feel about the current lunch app		
2. Find out why the library checkout line is slow by watching it		
3. See whether first-time users can figure out the new menu		
4. Understand in depth what the school nurse needs from a health app		



Think:

Signal words! “feedback,” “after testing,” or “revised” → iterative. “First... then... then...” with each piece working → incremental.

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Exam Practice

Part 8 · Exam Practice

★ Warm-up

1. Which is an example of **incremental** development?
- A. Rewriting the whole program after every test
 - B. Writing all of the code before doing any testing
 - C. Building a game one level at a time, testing each level before adding the next
 - D. Adding comments after the program is finished

★★ On Level

2. A team surveys 200 students and observes the cafeteria line before writing any code. This is part of which phase?
- A. Investigation
 - B. Prototyping
 - C. Testing
 - D. Debugging

★★ On Level

3. Why should developers document **throughout** development instead of only at the end?
- A. Comments make the code run faster
 - B. The computer needs comments to execute the program
 - C. Documentation is only needed for team projects
 - D. It helps the developer and teammates understand and maintain code as it is built

★★★ Exam Hard

4. A student uses an image-filter procedure written by a classmate in their Create task. Which action is required?
- A. Change the procedure's name so it looks original
 - B. Acknowledge the classmate as the author in the program
 - C. Nothing, since classmates are allowed to share code
 - D. Delete the procedure and start over

★★★ Exam Hard

5. Which **TWO** statements about program documentation are true? (*Select two.*)
- A. Documentation is only needed for programs written by teams.
 - B. Comments do not affect how a program runs.
 - C. Every programming environment supports comments.
 - D. Documentation helps in developing and maintaining correct programs.

Create Task Connection

Think about the program you will build for your Create task. Describe (a) one investigation method you'd use and why, (b) the first increment you'd build, (c) how you'd use feedback iteratively, and (d) how you'd document and credit any outside code.
